

Example 1 : From the single proposition $p \wedge (p \rightarrow q)$ show the q is a conclusion.

Solution:-

Step 1 : $p \wedge (p \rightarrow q)$ [premise]

Step 2 : p [1, simplification]

Step 3 : $p \rightarrow q$ [1, simplification]

Step 4 : q [2, 3, Modus ponens]

Example 2 : 1. $T \rightarrow (M \vee E)$

2. $S \rightarrow \neg E$

3. $T \wedge S$

show that M is a conclusion

Solution : Step - 1 : $T \rightarrow (M \vee E)$ [Premise]

Step - 2 : $S \rightarrow \neg E$ [premise]

Step - 3 : $T \wedge S$ [premise]

Step - 4 : T [3, Simplification]

Step - 5 : S [3, Simplification]

অথবা প্র্যান্ড এর ক্ষেত্রে P অথবা Q
 যা Q অথবা P বাক্য

Step-6 : $M \vee E$ [14, Modus Ponens]

Step-7 : $\neg E$ [2, 5, Modus Ponens]

Step-8 : M [6, 7 Disjunctive syllogism]

Example-03 :- Show the Hypothesis.

"It is not sunny this afternoon and it is colder than yesterday"

"We will go swimming only if it is sunny".

"If we don't go swimming then we will take a ^{canoes} ~~cars~~ trip"

"If we take a ^{canoes} trip, then we will be here ^{trip} by sunset"

Solution :-

P : It is sunny this afternoon.

Q : It is colder than yesterday.

R : we will go swimming.

S : we will take a ^{canoes} trip.

T : we will be at home by sunset.

Step 1 : $\neg P \wedge Q$ [Premise]

Step 2 : $R \rightarrow P$ [Premise]

Step 3 : $\neg R \rightarrow S$ [Premise]

Step-4 : $S \rightarrow T$ [Premise]

Step 9 10 11

Step - 5 : πP [1, Simplification]

Step - 6 : πn [2, 5, Modus Tollens]

Step - 7 : S [3, 6 Modus Ponens]

Step - 8 : T [4, 7 Modus Ponens]

"Homework" $\rightarrow$ 2020 @ 3 @ 6

2018 u 2 u 6

- [premise]
- [premise]
- [premise]
- [premise]

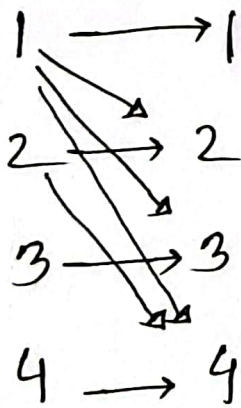
- Step 1 : $p \vee q$
- Step 2 : $p \rightarrow q$
- Step 3 : $\neg p \rightarrow \neg q$
- Step 4 : T

Example - 04 :- Let A be the set $\{1, 2, 3, 4\}$ which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution :-

Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that...

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$



R	1	2	3	4
1	x	x	x	x
2		x		x
3			x	
4				x

Displaying the ordered pairs in the Relation R .

Example - 08 :-

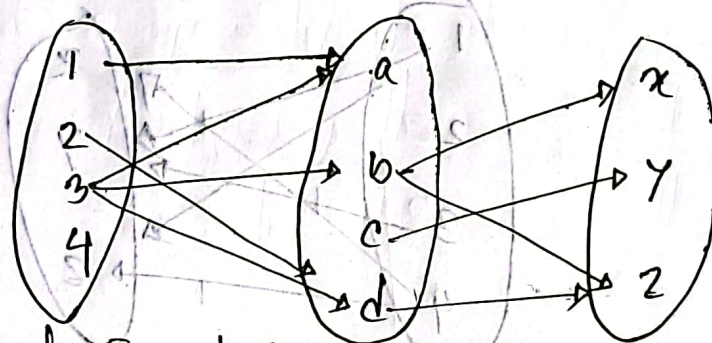
Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$ and

Let $R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\}$ and

$S = \{(b, x), (b, z), (c, y), (c, z)\}$ Consider the arrow

Diagram of R and S.

Solution :-



$\therefore$ Arrow Diagram of R and $S = R \circ S$.

$\therefore R \circ S = \{(2, z), (3, x), (3, z)\}$.

Example 10 :- Given $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$.

Let R be the following relation from A to B .

$R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$.

(a) Determine the matrix of the relation.

(b) Draw the arrow diagram of R .

(c) Find the inverse relation R^{-1} of R .

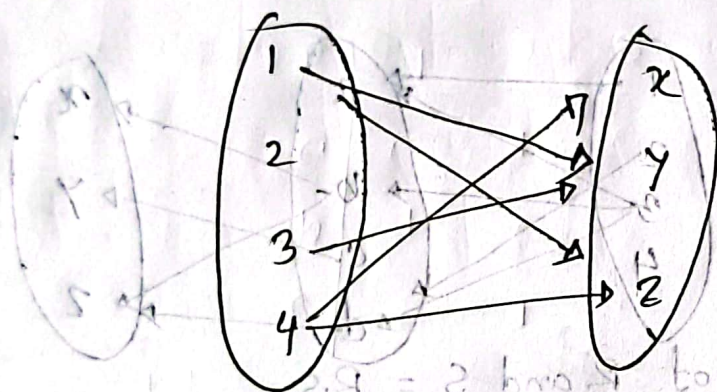
(d) determine the domain & Range of R.

Solution

(a)

	x	y	z
1	0	1	1
2	0	0	0
3	0	1	0
4	1	0	1

(b)



(c) $R^{-1} = \{(y, 1), (z, 1), (y, 3), (x, 4), (z, 4)\}$

(d)

Domain (R) = $\{1, 2, 3, 4\}$

Range (R) = $\{x, y, z\}$

Example 11 :- Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $C = \{x, y, z\}$,
 $R = \{(1, b), (2, a), (2, c)\}$; $S = \{(a, y), (b, x), (c, y), (c, z)\}$.

① Find the composition relation $R \circ S$.

② Find the matrices M_R , M_S and $M_{R \circ S}$.

Solution : $R \circ S = \{(1, x), (2, y), (2, z)\}$.

Example 12 :- Consider the relation $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$ on $A = \{1, 2, 3, 4\}$.

① Find $R^2 = R \circ R$.

② Draw the directed graph/digraph.

Solution :-

$$R^2 = R \circ R$$

$$= \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 3), (4, 4)\}$$

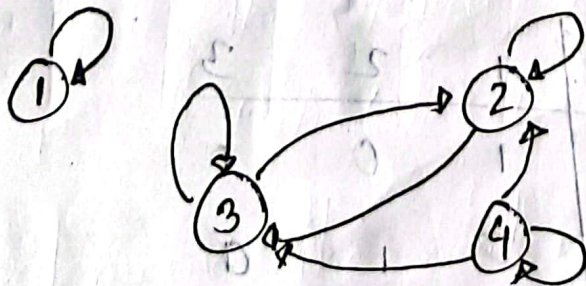
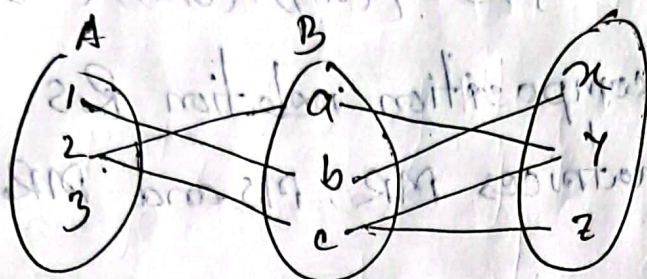


Figure ① : Directed graph of R^2 .

Solution 4:

a.



$$R.S = \{(1, x), (2, y), (2, z)\}$$

M.R

R	1	2	3
a	0	1	0
b	1	0	0
c	0	1	0

M.S

S	a	b	c
x	0	1	0
y	1	0	0
z	0	0	1

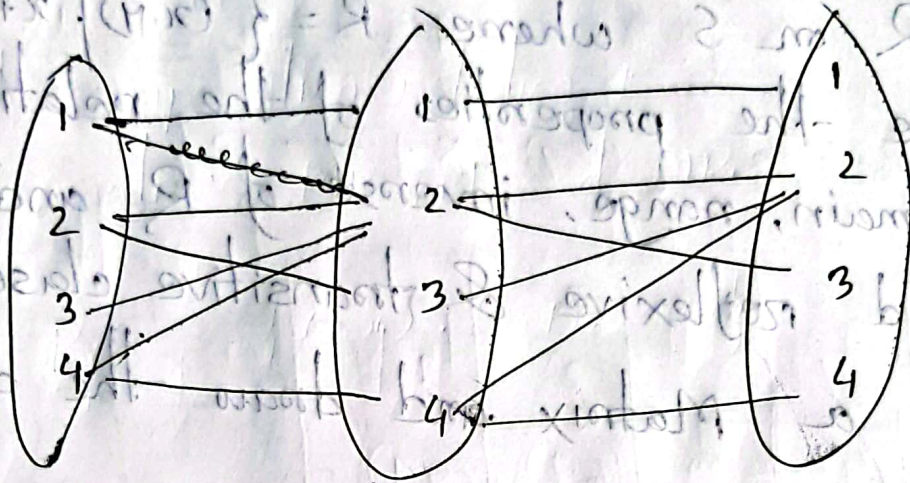
M.R.S

R.S.	1	2	3
x	1	0	0
y	0	1	0
z	0	0	1

Q. 2 on 9/1/21

Example 12: $R = \{(1,1), (2,2), (2,3), (3,2), (4,2), (4,4)\}$.

Solution:



$$\therefore R^2 = R \circ R = \{(1,1), (2,2), (2,3), (3,2), \underline{(3,3)}, (4,2), (4,4)\}.$$

	e	a	b	c	d	e	f	g	h
e	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	(1,7)	(1,8)	(1,9)
a	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)	(2,7)	(2,8)	(2,9)
b	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)	(3,7)	(3,8)	(3,9)
c	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)	(4,7)	(4,8)	(4,9)
d	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)	(5,7)	(5,8)	(5,9)
e	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)	(6,7)	(6,8)	(6,9)
f	(7,1)	(7,2)	(7,3)	(7,4)	(7,5)	(7,6)	(7,7)	(7,8)	(7,9)
g	(8,1)	(8,2)	(8,3)	(8,4)	(8,5)	(8,6)	(8,7)	(8,8)	(8,9)
h	(9,1)	(9,2)	(9,3)	(9,4)	(9,5)	(9,6)	(9,7)	(9,8)	(9,9)

Like as
2019 (5.10)

1. Given $S = \{1, 2, 3, 4, \dots, 10\}$, and a Relation R on S where $R = \{(x, y) \mid x + y = 10\}$, what are the properties of the relation R ? Find Domain, range, inverse of R and Find $R \circ R$. Find reflexive & transitive class. Replace R as a Matrix and draw the diagram of R .

Solution:- $\{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1)\} = R \circ R = R$

$$R = \{(x, y) \mid x + y = 10\}$$

Given, $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

when, $R = \{(x, y) \mid x + y = 10\}$.

x	1	2	3	4	5	6	7	8	9
y	9	8	7	6	5	4	3	2	1
(x, y)	(1, 9)	(2, 8)	(3, 7)	(4, 6)	(5, 5)	(6, 4)	(7, 3)	(8, 2)	(9, 1)

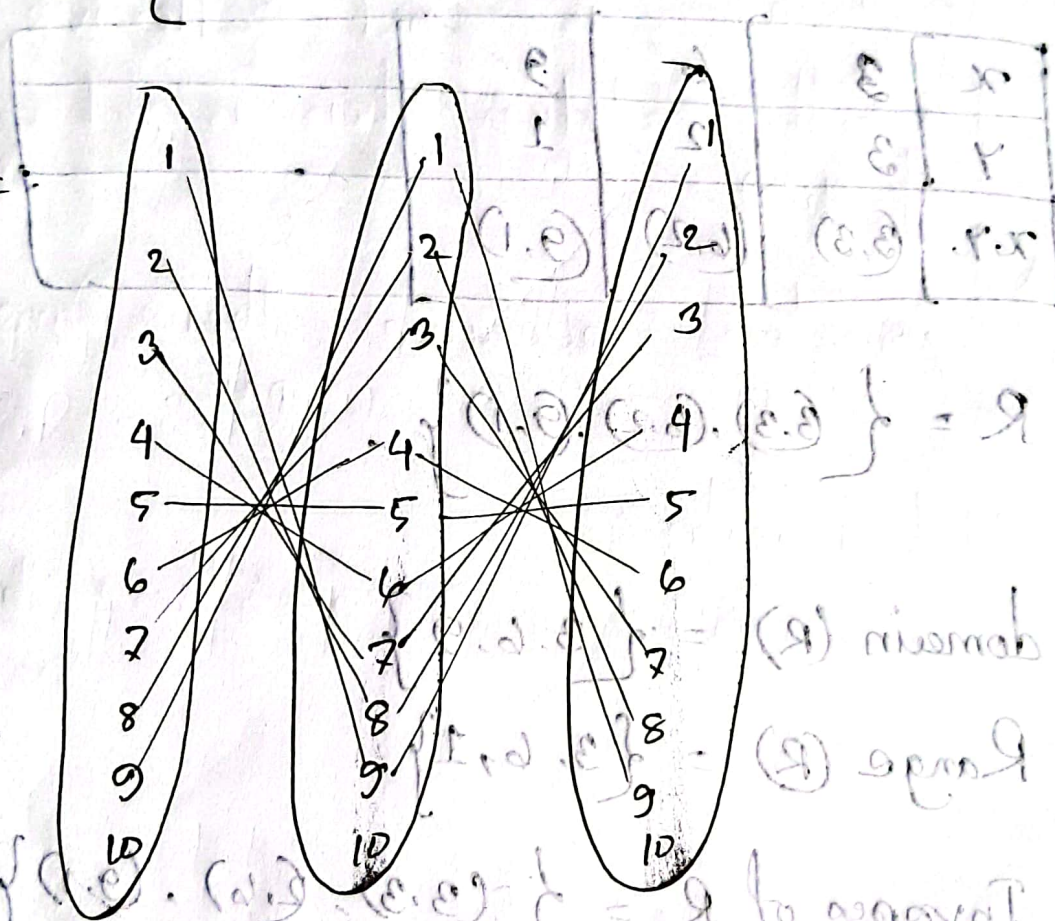
$$\therefore R = \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1)\}$$

$$\text{Domain } (R) = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

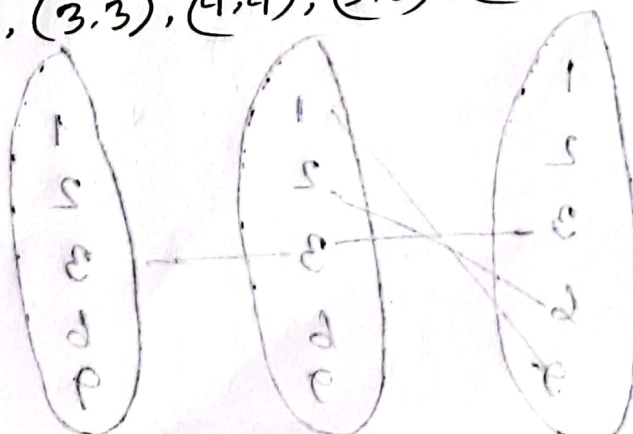
$$\text{Range } (R) = \{9, 8, 7, 6, 5, 4, 3, 2, 1\}$$

$$R^{-1} = \{(9,1), (8,2), (7,3), (6,4), (5,5), (4,6), (3,7), (2,8), (1,9)\}$$

$R \circ R =$



$$\therefore R \circ R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (7,7), (8,8), (9,9)\}$$



2019

5-6

let,

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

when, $R = \{(x, y) : x + 3y = 12\}$

x	3	6	9
y	3	2	1
x, y	(3, 3)	(6, 2)	(9, 1)

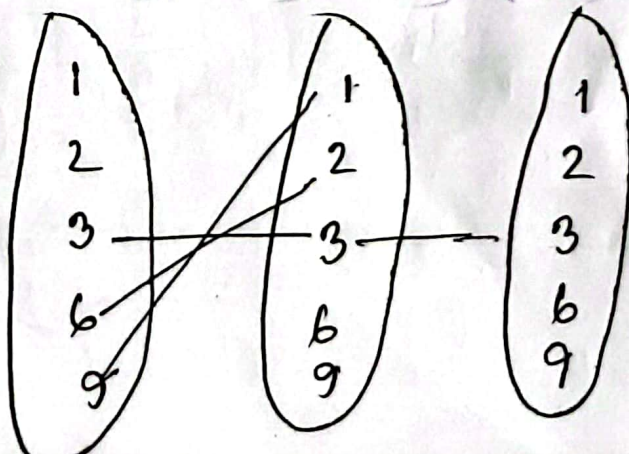
① $R = \{(3, 3), (6, 2), (9, 1)\}$

② domain (R) = $\{3, 6, 9\}$

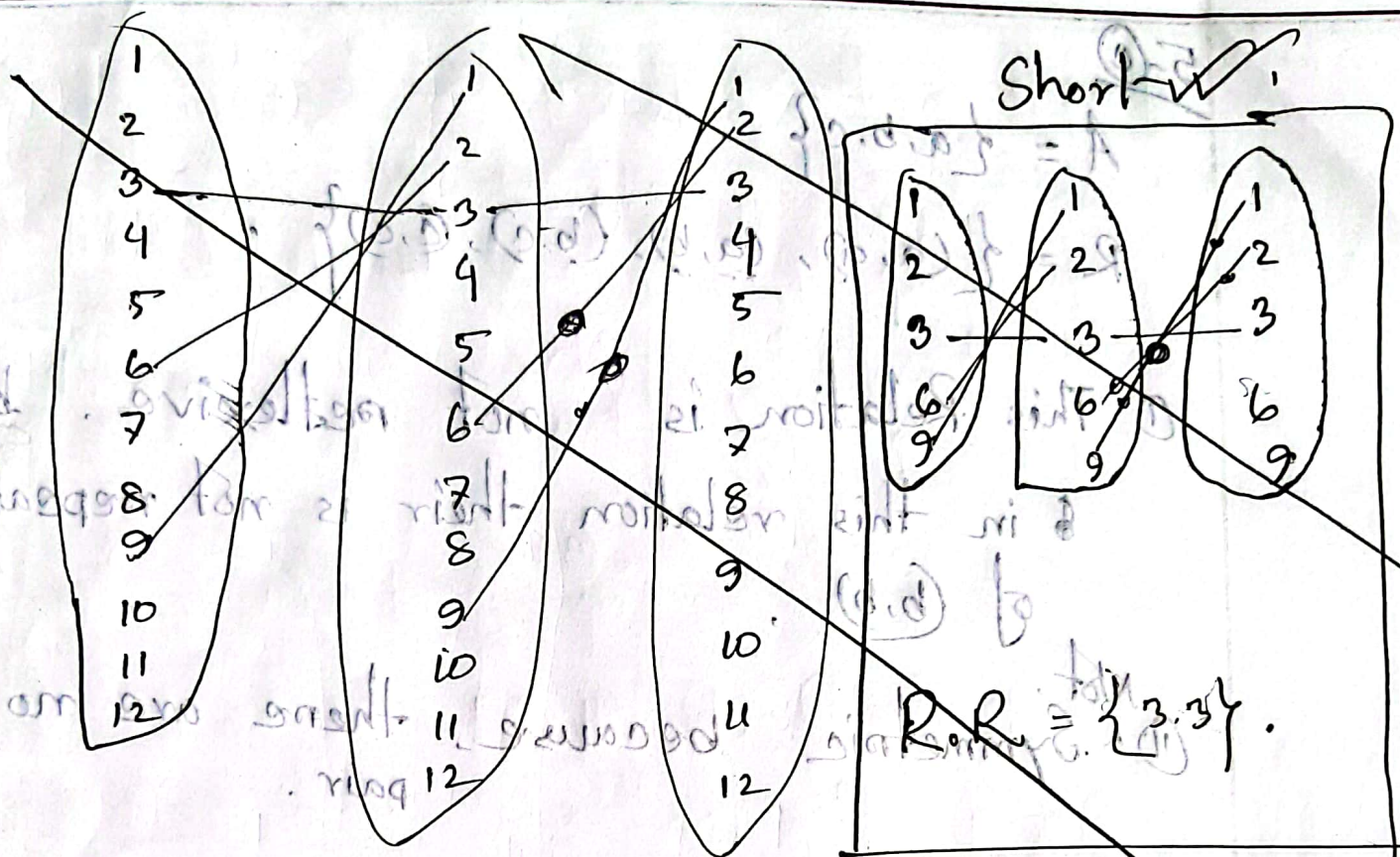
Range (R) = $\{3, 2, 1\}$

Inverse of R = $\{(3, 3), (6, 2), (9, 1)\}$

③



$\therefore R.R = \{(3, 3)\}$



$$R.R = \{(3, 3)\}.$$

5-c

$$A = \{a, b, c\}$$

$$R = \{(a, a), (a, b), (b, c), (c, c)\}$$

① This Relation is not reflexive, because in this relation there is not repeat pair of (b, b) .

② Not Symmetric because there are no inverse pair.

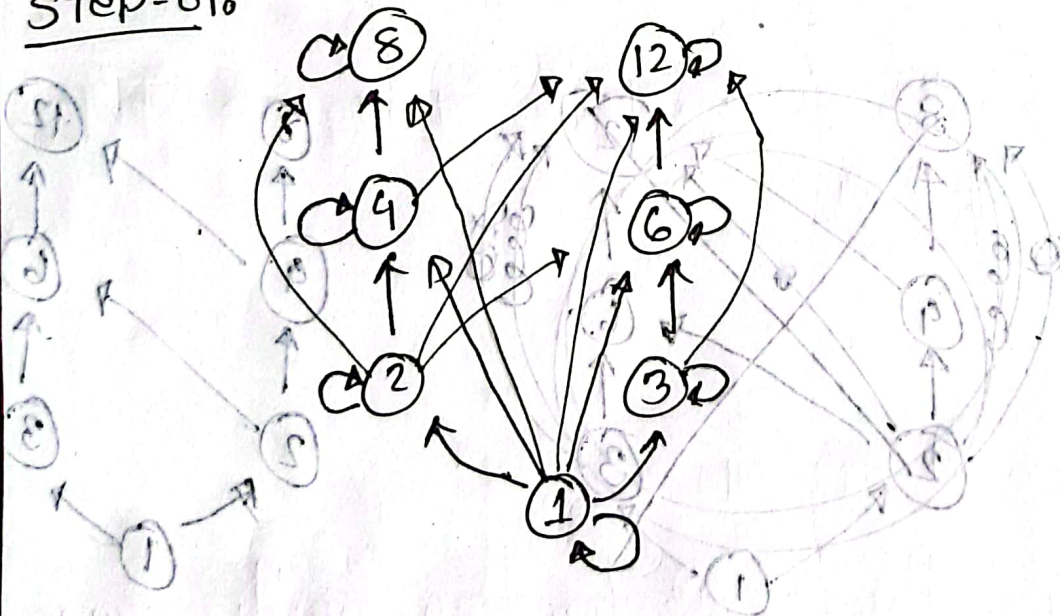
③ Not transitive
 $(a, a), (a, b)$ are there but (b, a) not here.

Write the steps & draw the Hasse diagram representing the partial ordering $\{(a,b) | a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.

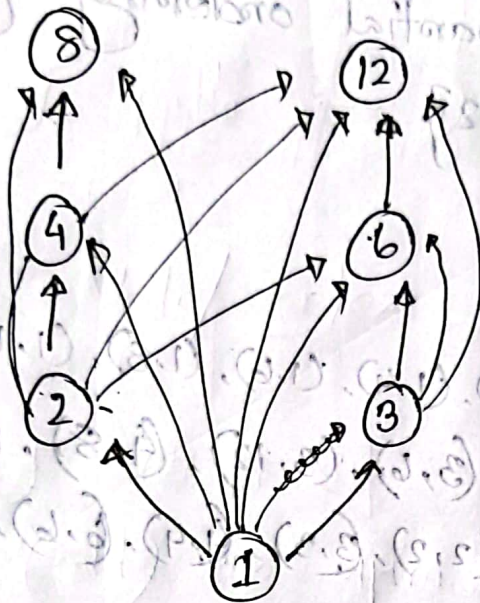
Answer:-

$$R = \{ (1,2), (1,3), (1,4), (1,6), (1,8), (1,12), (2,4), (2,6), (2,8), (2,12), (3,6), (3,12), (4,8), (4,12), (6,12), (1,1), (2,2), (3,3), (4,4), (6,6), (8,8), (12,12) \}$$

step-01:-

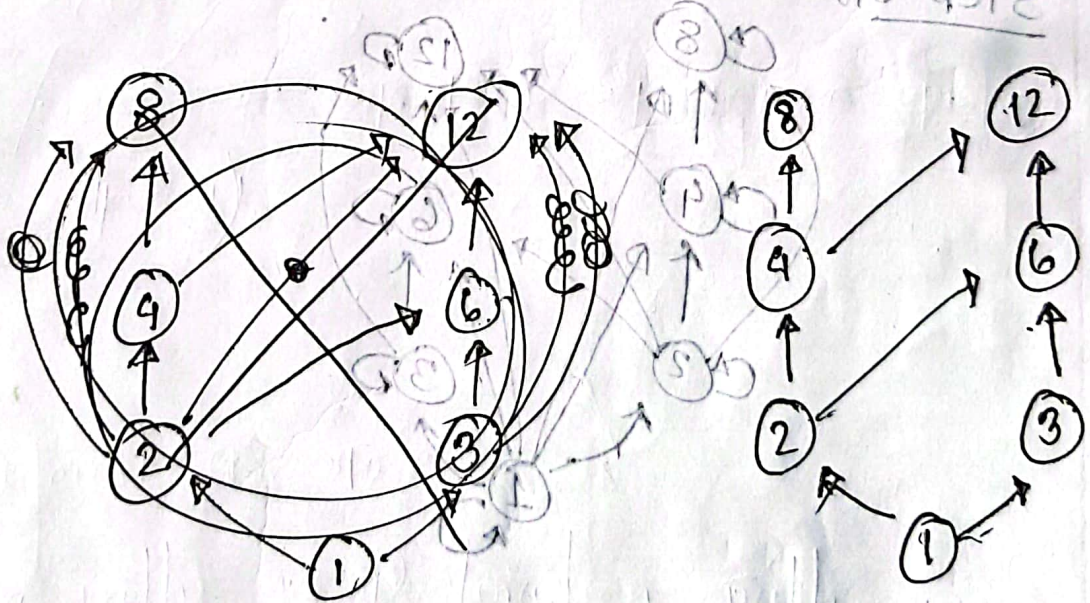


Step-02:



Step-02:
repeat pair बाद:

Step-03:



नॉट transitive करतो शक -

= (a,b) शाकल (b,c) शाकल (a,c) शाकल
याबे ना

Isomorphic graph (এনার উদাহরণ ৬)

- ① দু'টি graph এর vertex / Node সমান হতে হবে।
- ② দু'টি graph এর E edge সংখ্যা সমান হতে হবে।
- ③ দু'টি graph এর E Degree সমান হতে থাকতে হবে।
- ④ Degree of sequence should be same.

(Isomorphic graph চিত্রে ২(এ) দুই বা এর
আরেক graph এর সাথে তুলনা করা হয়)

নিম্নের দু'টি গ্রাফের মধ্যে কোন গ্রাফের কোন গ্রাফের

সংখ্যা বা ক্রমিক ১০ করে গ্রাফের কোন গ্রাফের

৬ এর ১০ গ্রাফের মধ্যে

১ গ্রাফের মধ্যে ১০ গ্রাফের মধ্যে ১০ গ্রাফের মধ্যে ①

১০ গ্রাফের মধ্যে ১০ গ্রাফের মধ্যে ১০ গ্রাফের মধ্যে ②

১ গ্রাফের মধ্যে

Euler Path চম্ভার উদ্যায়

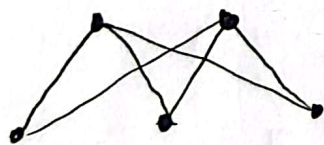
- ① Starting path এবং Destination/Ending path সমানদ্য হার।
- ② প্রত্যেকটি Node/vertex এর জোড়/even সংখ্যক Degree থাকতে হার।
- ③ এর ক্ষেত্রে যেকোন একটি vertex/node এর Degree বিজোড় (odd) হতে পারবে।
- ④ প্রত্যেকটি edge কে ভ্রমণ করতে হবে কিন্তু স্তূর্ধ্বিমাণ-একবার করেই।
(Node/vertex কে স্তূর্ধ্বিমাণ এক বা একাধিক বার স্তূর্ধ্বিমাণ করতে পারব।)

Euler circuit এর স্তূর্ধ্বিমাণ

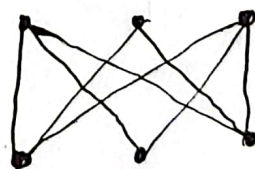
- ① Starting path এবং Destination path same হার।
- ② প্রত্যেকটি node/vertex এর জোড়/even সংখ্যক Degree থাকতে হার।

- ① Every vertex has even degree $\rightarrow$ circuit possible
- ② Exactly two vertices have odd degree $\rightarrow$ path possible / circuit not possible
- ③ More than two vertices have odd degree $\rightarrow$ path not possible / circuit not possible.

Complete Bipartite Graph :



$K_{2,3}$



$K_{3,3}$

Hamiltonian circuit / cycle : In a connected graph is defined as a closed walk that traverse every vertex of G exactly once, except of course the starting / ending vertex.

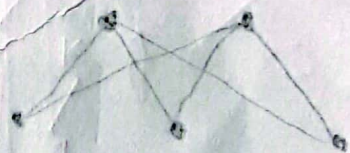
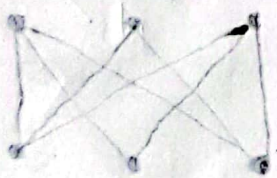
Hamiltonian graph :-

A graph G is said to be Hamiltonian, if it has a Hamiltonian circuit.

① Every vertex has even degree
 ② Exactly two vertices have odd degree
 both possible / circuit not possible

③ Every vertex has odd degree
 both not possible / circuit not possible

Complete Bipartite Graph :-



Hamiltonian circuit / cycle :- In a connected graph, a closed walk that traverses every vertex of G exactly once, except the starting / ending vertex, is called a Hamiltonian circuit / cycle.